

NISHITH VADLAMUDI

Software Engineer | Distributed Systems | Cloud Infrastructure | Integrations

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SUMMARY

Software Engineer with nearly 3 years of experience building large scale distributed systems and object oriented Java back end services on AWS for a platform processing millions of daily events. Led failure handling that protects about 1,000 shipments a week, launched a new API version with zero client impact, and validated 4.7x peak traffic with zero errors. Experienced in design reviews, code reviews, documentation, and triage. Master of Science in Computer Science.

TECHNICAL SKILLS

Languages	Java, Python, TypeScript, JavaScript, SQL, Go, C++, HTML, CSS, Object Oriented Programming
Back End	Spring Boot, REST APIs, Microservices, API Design, Event Driven Architecture, JDBC, Hibernate
Distributed Systems	Caching, Message Queues, Idempotency, Retry and Backoff, Concurrency, Fault Tolerance
Cloud and Networking	AWS Lambda, ECS, Fargate, VPC, DNS, API Gateway, DynamoDB, SQS, S3, AWS CDK, Terraform
Data and Algorithms	Data Structures, Algorithms, System Design, MySQL, MS SQL Server, Redis, Query Optimization
Quality and Security	JUnit, Mockito, TDD, Code Review, Design Review, Load Testing, OAuth 2.0, JWT, Secure Coding
Operations	Git, CI/CD, Docker, Kubernetes, Canary Deployments, CloudWatch, Debugging, Incident Triage

EXPERIENCE

Software Development Engineer, Amazon

January 2025 to Present

Large scale fulfillment platform, distributed Java services and integration infrastructure

Bellevue, WA

- Led graceful failure handling with error aggregation and 8 close states in Java, TypeScript, AWS Lambda, and AWS CDK through design review and phased rollout, protecting about 1,000 shipments a week from stuck workflows.
- Debugged a new API version failing 100 percent of calls by applying root cause analysis to 9 routing, credential, and serialization defects in Java services on SQS and DynamoDB, launching with zero client impact.
- Built an idempotent provisioning CLI with DynamoDB configuration, bulk runs, and safe reruns, following infrastructure as code practices, replacing 30 minutes of error prone manual setup per site with under 1 minute for all sites.
- Designed a per API capacity model and full scale load test for AWS Lambda, ECS, and DynamoDB services using peak traffic modeling, validating 4.7x projected peak with zero errors across 12.7 million invocations.
- Engineered a JWT aware token cache in TypeScript on AWS Lambda and DynamoDB batch write retries using exponential backoff with jitter, cutting redundant auth calls up to 3x at a 99.97 percent hit rate and closing a silent data loss path.
- Automated deployment safety with AWS Fargate canaries, CloudWatch alarms, and AWS CDK deployment gates following continuous delivery, enabling automatic rollback in every production region and catching 5 release blockers via fault injection.

Software Engineer, ValueMomentum

June 2021 to July 2022

Enterprise web applications and microservices

Hyderabad, India

- Developed a full stack workforce analytics platform with Angular, TypeScript, Spring Boot, and REST APIs, applying SQL query optimization and advanced UI filters, cutting report generation time 50 percent.
- Migrated over 10,000 client files from remote desktop servers into MS SQL Server using Java and JDBC, following a phased, validated migration plan, completing the transfer without disruption.
- Streamlined record closure with Java scheduled jobs using cron expressions and SMTP email notifications, following batch processing practices, reducing manual effort 80 percent.
- Rearchitected query processing in Java Spring Boot microservices secured with JWT and role based access control, shipping via Docker, Kubernetes, and Azure DevOps CI/CD, improving query efficiency 40 percent.

PROJECTS

Othello Multiplayer Gaming Application

Angular, TypeScript, Spring Boot, WebSockets, MySQL, Docker

- Created a multiplayer web app with an Angular and TypeScript frontend and Spring Boot microservices over REST and WebSockets, using MySQL, Docker, and JWT with OTP, syncing validated moves in under 100 ms.
- Applied Test Driven Development with Jasmine and Karma and GitHub pull request reviews, reaching 95 percent code coverage and 100 percent move validation accuracy across 50+ commits.

Concrete Compressive Strength Prediction

Python, NumPy, Pandas, Scikit learn, TensorFlow, Keras

- Predicted concrete compressive strength from 1,030 mix samples in Python by comparing 5 Scikit learn regression models, with Random Forest reaching 0.90 R squared and cutting RMSE 51 percent versus linear regression.

EDUCATION

Master of Science in Computer Science, George Mason University

2022 to 2024

Coursework: Analysis of Algorithms, Software Architecture and Design, Secure Software Design, Machine Learning